1. Unified Big Data Architecture for retail.

- A retail organization can use a data lake architecture to combine data from websites, mobile apps, physical address and CRM systems.

-> This creates a 360 degree customer views and supports personalized recommendation and targeted marketing.

Proposed Architecture

a) Data sensors
b) Data Ingestions
c) Processing
d) Analytics and ML
e) Visualization.

2. Distributed Computing for Scientific Research Data.

-> Scientific research after procedures very large datasets that cannot be efficiently processed by a single computer.

-> Parallel Processing allows these computers to work simultaneously.

Advantages over centralized systems

→ Faster processing
→ Better scalability
→ Improposed resource utilization
→ Handles large datasets.

3. Fraud Detection framework for Digital Payments.

A fraud dection frame work shoud process transaction data in real time.

Payment transactions → streaming Platfrom → Real-time Processing ↵
mL fraud model
Alert. ← fraud score ← mL fraud model

Technologies :-

a) Apache kafka
b) Apache spark streaming
c) machine learning models
d) Database.

4. streaming and processing multimedia Data:

Multimedia data such as images, Audio and videa creates challengess because of its large size, high processing requirements, diffrent formates.

challenges :-

→ High storage requirements.

* slow data transfer
* large processing requirements
* Different file formats
* metadata management

Recommendation Technologies

* cloud object storage
* HDFS
* Apche spark
* kafka
* CDN

5. Disaster Recovery and Busines Continuity stratagy.

A cloud-based big data platforms should maitain continous operations even when failures occurs.

stratagy:-

1. store data in multiple locations.
2. Maintain primary and secondary systems
3. use automated systems.

Key mechanisms.

1. Reduandancy
2. Replication
3. Automated failures.

6. Data Quality and Data Governance.

Poor data quality reduces the reliability of analytical results.

Major Issues :-

Inconsistency, Duplications, missing values

Data generation stratagies.

1. Data validation rules
2. Data standarization
3. Missing value handling
4. Regular data quality audits.

7. Scalable Recommendation system for streaming services.

A streaming platform can analyze user behavior to recommended relevant content.

User Activity → kafka → Data lake → Spark processing → Recommendation models → Personalized Recommendations.

Data Analyzed.

1. viewing history
2. search history
3. watch time
4. Likes and ratings
5. Content preferences.

8. Role of cloud Computing in Big Data.

Cloud computing provides infrastructure and services for storing and processing large volume of data.

Advantages

→ Scalability

2. Flexibility
3. cost Efficiency
4. Global access
5. managed services.

9. Real-time Analytics for Transportation and Logistics.

A transportation company can collect real-time data from GPS devices, vehicles, sensors, traffic systems and fuel-monitoring systems.

GPS/sensors → kafka → stream processing → Analytics → Route optimization system → Data dashboard.

10. Ethical and legal challenges in big data.

Big data collection creates important ethical and legal concerns

challenges

a) Privacy Violations
b) Lack of informate construct
c) Data breaches
d) Lack of transparency

Recommended policies:

→ Collect only necessary data

→ Obtain clear user consent

→ Use encryption and access control

→ Perform regular security audits